

A Theory of Economic Slack

Pascal Michailat

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APPENDIX I.

Inflation and tightness dynamics at the zero lower bound

In this appendix, we derive and study the properties of the dynamical system describing the solution to the slackish business cycle model of chapter 14 at the zero lower bound. In normal times, the central bank sets a real interest rate, which separates the dynamics of inflation and tightness. Tightness can therefore be determined as in the fixed-inflation model of chapter 13, and inflation can be determined from the Phillips curve. At the zero lower bound, this separation disappears as the central bank loses control of the real interest rate. The nominal interest rate is stuck at 0 so the real interest rate is just negative inflation: $r = i - \pi = -\pi$. Tightness and inflation are therefore jointly determined by a two-dimensional dynamical system.

I.1. Relationships between inflation, tightness, and the costate variables

The first step in building the two-dimensional dynamical system describing the model of chapter 14 at the zero lower bound is to express the costate variables ψ^b and ψ^p as a function of inflation π and tightness θ . Using these relationships, we will be able to transform the differential equations for the costate variables into differential equations for inflation and tightness.

This is easy to do for the costate variable ψ^p , as the relationship is directly given by

equation (14.16):

$$(I.1) \quad \psi^p = \omega [\pi - \pi^n]$$

$$(I.2) \quad \dot{\psi}^p = \omega \dot{\pi}.$$

For the costate variable ψ^b , the relationship arises from equation (14.15). Indeed, in the model, aggregate consumption is directly determined by tightness:

$$(I.3) \quad c(\theta) = \frac{y(\theta)}{1 + \tau(\theta)} = \frac{1 - u(\theta)}{1 + \tau(\theta)} \cdot kh.$$

Using the function $c(\theta)$, we can relate ψ^b to tightness using (14.15):

$$(I.4) \quad \psi^b = (1 - \alpha) \cdot \frac{c(\theta)^{-\alpha}}{1 + \tau(\theta)}.$$

Notice also, which will be useful later, that the product $\psi^b(\theta)y(\theta)$ is simply given by:

$$(I.5) \quad \psi^b(\theta)y(\theta) = \psi^b(\theta)c(\theta)[1 + \tau(\theta)] = (1 - \alpha)c(\theta)^{1-\alpha}.$$

Using a basic chain rule, the time derivative of the costate variable therefore is:

$$\dot{\psi}^b = \frac{\psi^b}{\theta} \cdot \epsilon_{\theta}^{\psi^b} \cdot \dot{\theta},$$

where $\epsilon_{\theta}^{\psi^b}$ is the elasticity of the costate ψ^b with respect to tightness θ .

To compute that elasticity (and others below), we use the result that the elasticity of $1 + \tau(\theta)$ with respect to θ is given by (6.12):

$$(I.6) \quad \epsilon_{\theta}^{1+\tau} = \eta(\theta)\tau(\theta).$$

Another useful elasticity is the elasticity of $1 - u(\theta)$ with respect to θ , which can be inferred from (13.13):

$$(I.7) \quad \epsilon_{\theta}^{1-u} = \frac{-u(\theta)}{1 - u(\theta)} \cdot \epsilon_{\theta}^u = [1 - \eta(\theta)] u(\theta).$$

From these results and (I.3), we infer that the elasticity of $c(\theta)$ with respect to θ is

$$\epsilon_{\theta}^c = [1 - \eta(\theta)] u(\theta) - \eta(\theta)\tau(\theta).$$

Then, from these results and (I.4), we infer that the elasticity of ψ^b with respect to θ is

$$\epsilon_{\theta}^{\psi^b} = -[\alpha[1 - \eta(\theta)]u(\theta) + (1 - \alpha)\eta(\theta)\tau(\theta)].$$

And finally from this we relate the time derivative of the costate variable ψ^b to that of tightness θ :

$$(I.8) \quad \frac{\dot{\psi}^b}{\psi^b} = -[\alpha[1 - \eta(\theta)]u(\theta) + (1 - \alpha)\eta(\theta)\tau(\theta)] \cdot \frac{\dot{\theta}}{\theta}.$$

I.2. Differential equation for inflation

Using the results from the previous section, we now transform the differential equation for the costate variable ψ^p , given by (14.18), into a differential equation for inflation π . We simply replace ψ^p and $\dot{\psi}^p$ in the equation by the expressions (I.1) and (I.2), and $\psi^b y$ by the expression (I.5). We obtain:

$$(I.9) \quad \dot{\pi} = \delta[\pi - \pi^n] + \frac{1 - \alpha}{\omega} \cdot c(\theta)^{1 - \alpha} \left[1 - \frac{\eta(\theta)}{1 - \eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)} \right].$$

Accordingly, the nullcline $\dot{\pi} = 0$ is given by:

$$(I.10) \quad \pi = \pi^n + \frac{1 - \alpha}{\delta \omega} \cdot c(\theta)^{1 - \alpha} \left[\frac{\eta(\theta)}{1 - \eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)} - 1 \right].$$

This expression is close but not the same as the Phillips curve in normal times, given by equation (14.20). This is because the costate variable ψ^b is fixed, determined by the real rate in normal times, whereas it is jointly determined by tightness and inflation at the zero lower bound.

I.3. Differential equation for tightness

Next, we transform the differential equation for the costate variable ψ^b , given by (14.17), into a differential equation for tightness θ . First, we divide the equation by ψ^b . We also set the real interest rate to $r = -\pi$. We obtain

$$\frac{\dot{\psi}^b}{\psi^b} = (\delta + \pi) - \frac{1}{a\psi^b}.$$

Next, we replace ψ^b and $\dot{\psi}^b$ in the equation by the expressions (I.4) and (I.8). This yields:

$$(I.11) \quad [\alpha[1 - \eta(\theta)]u(\theta) + (1 - \alpha)\eta(\theta)\tau(\theta)] \cdot \frac{\dot{\theta}}{\theta} = \frac{1}{a(1 - \alpha)} \cdot [1 + \tau(\theta)]c(\theta)^\alpha - (\delta + \pi).$$

From this equation, we see that the nullcline $\dot{\theta} = 0$ is given by:

$$(I.12) \quad \pi = \frac{1}{a(1-\alpha)} \cdot [1 + \tau(\theta)] c(\theta)^\alpha - \delta.$$